

FastAPI Agent Service

REST routes | Pydantic schemas | OpenAPI docs | request/response interface

1. External Workflow

Goal Manager

Strategy Registry

Strategy Planner

Safety Guard

Autonomous Experiment Orchestrator

2. API Schema & Adapter

StrategySchema

RegisterStrategyRequest

GenericStrategyAdapter

3. Controller/safety

diagnostics → ordered state vector

ControllerFactory

*Creates*

Classical  
Agent

Quantum  
Agent

Hybrid  
Agent

Deep-Unfolded  
Agent

*implemented by*

Deep-Unfolded Controller Backend  
state-to-control mapping, differentiable update logic,  
adaptive policy, learnable parameters

bounded  
controls

bounded  
controls

4. Executor/  
Logging

Classical /  
Federated Executor

Quantum /  
Qiskit Executor

Hybrid  
Executor

External HTTP  
Executor

Continue/  
Stop/Switch

ExperimentLogger  
metrics, events, traces,  
controller metadata

Memory /  
SQLite Persistence  
controllers, metadata,  
state, metrics

Checkpoint Manager  
controller checkpoints,  
snapshots

Evaluator  
performance analysis,  
scoring, comparison

5) Persistence &  
Monitoring Layer

Run Summary + Best Result  
final artefact (summary report, best controller,  
metrics)

Internal controller service flow

Pipeline 1: Controller creation

POST /controllers/create

ControllerFactory

controller instance

Pipeline 2: Suggest controls

POST /controllers/{id}/suggest-controls

GenericStrategyAdapter

Deep-Unfolded Controller Backend

bounded controls

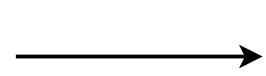
Pipeline 3: Controller update

POST /controllers/{id}/update-controller

feedback (metrics, rewards)

logger / SQLite storage  
(persistence & checkpoints)

Containerised deployment and  
reproducible execution supported via Docker.



Primary internal control flow



Schema / data flow (structures, diagnostics)



Controller update flow (models, parameters)



Persistence / logging flow (metrics, checkpoints)